

HJARTASEN, Norway

WMO: 011500

Lat: 66.499N

Lon: 14.954E

Elev: 823

StdP: 14.26

Time Zone: 1.00 (EUC)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.3	-2.5	-11.6	3.0	-4.0	-8.6	3.6	-1.1	22.9	30.7	18.8	27.1	3.3	90	0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.7	76.8	59.8	72.1	58.3	67.8	55.7	62.2	73.0	59.8	69.0	57.6	65.6	5.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.3	75.0	65.2	55.7	68.1	62.3	53.6	63.1	60.9	28.3	73.6	26.6	69.1	25.2	65.5	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	14.0	11.9	-10.8	81.9	4.5	5.3	-14.0	85.7	-16.7	88.9	-19.2	91.8	-22.5	95.7		
(5)			-11.3	64.6	4.5	3.8	-14.5	67.3	-17.2	69.6	-19.7	71.7	-22.9	74.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	18.3	20.8	24.4	33.1	42.1	49.6	56.8	53.8	47.5	36.2	26.9	23.6
	DBStd	15.35	11.26	10.93	9.35	6.49	6.68	5.52	6.43	5.08	5.83	6.87	9.92	10.11
	HDD50	5517	984	817	794	507	263	73	8	20	112	430	692	817
	HDD65	10535	1449	1237	1259	957	710	463	267	350	525	894	1142	1282
	CDD50	475	0	0	0	0	18	61	221	136	38	1	0	0
	CDD65	16	0	0	0	0	0	0	14	2	0	0	0	0
	CDH74	248	0	0	0	0	10	6	213	19	0	0	0	0
	CDH80	41	0	0	0	0	1	0	40	0	0	0	0	0
	WSAvg	4.8	4.9	5.6	5.5	5.0	5.2	5.0	4.1	4.0	4.2	4.7	4.8	4.9
	PrecAvg	49.4	6.1	5.4	4.9	2.2	2.3	2.9	3.4	3.5	4.0	4.9	3.7	5.7
(15) Precipitation	PrecMax	59.0	12.2	11.3	10.2	4.9	3.5	4.8	7.9	5.8	9.0	12.2	9.8	13.2
	PrecMin	39.0	0.9	0.7	0.9	1.0	1.3	0.7	0.8	1.5	0.9	0.7	0.4	1.8
	PrecStd	6.5	3.6	2.9	2.9	1.2	0.6	1.2	1.6	1.2	2.2	2.7	2.8	3.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.5	42.0	43.6	53.8	74.3	74.5	84.4	76.7	68.0	54.1	47.0	42.3
		MCWB	39.7	38.3	37.0	43.1	54.0	57.5	64.8	60.1	55.3	46.8	43.1	39.1
	2%	DB	37.5	39.6	41.1	48.4	65.0	68.6	79.9	72.8	62.1	50.5	44.9	40.0
		MCWB	34.5	36.7	36.6	38.9	50.6	53.9	62.1	58.9	52.7	45.6	42.1	37.1
	5%	DB	34.6	37.5	38.9	44.4	60.1	65.1	75.6	68.4	59.1	47.9	41.2	37.5
		MCWB	32.5	34.5	34.9	37.0	48.3	51.9	60.5	57.4	51.7	43.6	38.3	34.8
	10%	DB	32.6	35.3	36.5	41.7	54.0	61.5	70.4	64.3	56.5	45.5	38.0	35.2
		MCWB	31.0	32.7	33.1	36.0	45.3	50.6	58.0	55.4	50.7	42.4	35.5	33.1
	0.4%	WB	39.8	39.3	39.4	43.3	54.5	58.5	67.8	63.2	57.4	48.5	44.8	40.3
		MCDB	41.3	41.2	42.0	53.0	72.1	68.7	80.2	70.7	63.9	52.3	46.3	41.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.2	37.3	37.2	40.3	51.3	56.1	64.3	60.4	54.8	45.9	41.9	37.5
		MCDB	36.7	39.5	40.1	45.6	63.9	65.2	75.3	68.1	60.2	49.4	44.6	39.6
	5%	WB	32.8	34.9	35.2	38.4	48.5	54.0	61.6	58.5	52.9	44.3	38.9	35.0
		MCDB	34.2	37.0	38.5	42.7	58.4	62.4	72.8	66.9	57.5	47.3	41.3	36.8
	10%	WB	31.1	32.9	33.3	36.7	45.8	51.6	59.4	56.4	51.1	42.8	35.8	33.3
		MCDB	32.5	34.9	36.0	40.7	53.4	59.5	68.5	63.3	55.4	45.4	37.7	34.8
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	12.0	14.7	14.7	15.2	15.7	17.7	16.8	14.2	9.7	8.9	9.8
		MCDBR	9.4	10.1	12.0	17.5	25.1	25.1	28.6	25.7	19.3	10.7	8.8	9.8
		MCWBR	8.1	8.0	8.7	10.4	12.7	12.3	12.7	13.4	11.1	7.5	7.2	7.8
	5% WB	MCDBR	9.4	10.0	10.8	14.5	23.4	21.5	24.0	22.7	16.8	9.0	8.6	9.8
		MCWBR	8.4	8.4	8.3	8.9	12.3	10.9	11.3	12.7	10.3	6.7	7.0	8.0
(40) Clear-Sky Solar Irradiance	taub		0.192	0.259	0.297	0.323	0.333	0.333	0.344	0.330	0.304	0.272	0.197	0.197
	taud		1.911	2.102	2.134	2.159	2.244	2.370	2.422	2.463	2.509	2.272	1.959	1.923
	Ebn at Noon		66	185	238	263	274	278	270	263	244	181	60	5
	Edh at Noon		5	17	28	35	36	33	30	26	20	14	4	1
(44) All-Sky Solar Radiation	RadAvg		20	140	556	1160	1551	1565	1417	1071	563	198	44	2
	RadStd		2	27	81	95	87	139	168	115	64	34	4	0

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page